

SereneCare: A Secure, Full-Stack Healthcare Platform Built Using Open Source Tools

SereneCare, a healthcare management platform designed to bridge the gap between patients, doctors, and administrators, is focused on creating a unified, full-stack system that supports appointment scheduling, group consultations, health resource sharing, and role-based workflows. Built entirely using open source technologies, SereneCare aims to offer a modular, scalable, and secure solution for modern healthcare challenges.

As developers exploring the intersection of healthcare and open source technology, we noticed a common thread: many existing platforms were either too specialised or lacked the flexibility to adapt to varied medical workflows. With that in mind, we set out to build SereneCare, a healthcare management system that prioritises usability, modularity, and data privacy.

The platform was conceptualised with real-world challenges in mind—limited access to in-person consultations, the need for support group integration, and growing concerns over the security of patient data. SereneCare was our attempt to create something that could support remote healthcare delivery without becoming overly complex for either patients or medical staff.

From the ground up, we built the system using tools we were familiar with—React for the frontend, Node.js and Express for the backend, and MongoDB to store flexible, evolving data structures. But beyond the tech stack, what really drove the project was our desire to build something meaningful: a platform that could be used in universities, clinics, or even small NGOs running wellness programmes. Every feature—from appointment management to real-time group discussions—was designed with adaptability in mind.



Role-based user interface design

Admin dashboard: For admins, we focused on functionality that allows them to control and monitor the entire system. The dashboard presents metrics like the number of registered users, pending approvals, and active support groups. Admins can easily verify new doctor signups, approve users, assign moderators to groups, and keep an eye on activity across the platform.

An admin-facing interface summarises system status and allows for user/group management in just a few clicks.

Doctor dashboard: Doctors needed more than just a schedule; they also needed a way to engage with patients outside of direct consultations. So, we built them a dashboard where they can:

- Review and manage appointment requests
- Moderate the support groups they're assigned to
- Share articles or documents with their patients
- Track their availability and upcoming sessions

The design borrows from minimalist medical record tools; so it feels familiar and focused.

This panel allows doctors to oversee appointments, engage in support groups, and contribute educational resources.

Patient dashboard: The patient interface was where we invested a lot of time. We wanted it to feel like a personal assistant for managing health, and not just a generic portal. Patients can:

- Schedule or reschedule appointments
- Join support groups based on interest or condition
- Browse health documents shared by doctors
- Ask questions and bookmark content for later

The layout was tested among peers to make sure it worked well across devices, especially phones and tablets.

Patients get a streamlined dashboard where they can interact with doctors, track appointments, and access wellness materials.

What made this part interesting for us as developers was the way React